

1. Understanding risk and return: an overview

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INTRODUCTION

Investment risk and return include a wide set of conceptual ideas and mathematical terms used to calculate the potential loss or gain from investing in tradable assets. Each asset class, specifically each security, can exhibit risk and return characteristics. Yet, one common characteristic remains the same. The risk and expected return for any particular investment are directly related. Thus, the greater the risk for an investment, the higher the potential for profit or loss. Note that this basic investing principle does not automatically imply that the higher the risk, the higher the return. As tempting as it might be to associate “return” with good outcomes and “risk” with bad outcomes, this assertion is technically a misnomer. The outcome is the return, good or bad. In other words, return is a measure of investment gain or loss. Risk, in this equation, is simply a measure of uncertainty. However, this oversimplification of risk can lead to more uncertainty because uncertainty can apply to either positive or negative returns. Mathematically, this issue is solved by assigning upside risk as the uncertain possibility of gain and downside risk as the uncertain possibility of loss. In general, though, investment risk management focuses primarily on the downside. Thus, a more useful definition of investment risk is the probability of losses relative to an investment’s expected return.

Investment management can be simplified into: (1) identifying the critical risks for specific investments and portfolios, and (2) mitigating or managing those risks to meet investors’ return objectives and risk tolerance. Yet one major complication with the first step is the many investment risks to consider. As a case in point, Jorion (2007) highlights three risks investors face: liquidity, credit, and operational risks. Liquidity risk is associated with the inability to liquidate a position quickly or sufficiently at a fair market price. Credit risk accounts for the probability of default by a counterparty in a financial transaction. Finally, operational risk measures the risk of loss from various

inefficiencies or failures from different operational sources, including people, processes, systems, and external events. Therefore, some risks account for several layers of additional risks. Moreover, some risks, like operational risks, impact some investment products more than others. Consider complex investment products like highly leveraged derivatives. Even small operational errors can drastically alter investment returns.

The second step in the investment management process is not so simple either. Investors should understand and employ several risk and return measures depending on their specific risk and return preferences and time horizons. For instance, risk can be measured in several ways, including standard deviation and beta. Returns can also be measured in several ways depending on how the returns account for fees, adjust for time, and treat periodic payments such as dividends and interest. Although risk is an important variable in determining future expected returns, risk is often excluded from the calculations for past returns. One caveat is using risk-adjusted returns, which measure an investment's excess returns (i.e., the return above the risk-free rate) per unit of risk. Two of the most popular measures include Sharpe's ratio (Sharpe, 1966, 1994), which uses standard deviation to measure risk, and Treynor's ratio (Treynor, 1966), which uses beta to measure risk. The Sortino ratio (Sortino and Price, 1994) modifies the Sharpe ratio to measure risk-adjusted returns using only the downside standard deviation. Finally, another measure of downside risk is value at risk (VaR), which quantifies the extent of possible financial losses over a defined period, given both normal market conditions and a stated level of confidence. With practice, mathematically calculating some of these concepts becomes easy. Other concepts, such as behavioral factors like investors' risk tolerances, are very difficult to quantify. Nonetheless, investment management involves identifying, analyzing, and managing the risk and return factors that lead to optimal investment decision-making.

Fortunately, tools have advanced for optimizing investment portfolios. Markowitz (1952) was one of the first to mathematically examine assembling a portfolio of assets based on risk-return tradeoffs. This research led to Markowitz's Modern Portfolio Theory (MPT), or mean-variance analysis, which is a statistical framework for selecting a portfolio of investments such that the expected return is maximized for a given level of risk. Building upon this foundation, Sharpe (1964) and Lintner (1965) were the chief architects of the Capital Asset Pricing Model (CAPM), which is a one-factor model used to determine the appropriate expected return given an investment's beta. Beta is a systematic risk variable that measures an investment's sensitivity to market risk, the primary investment risk in a well-diversified portfolio. Taken a step further, investors can also use the CAPM to calculate Jensen's alpha (Jensen, 1967), which is the abnormal return between an investment's actual return and its theoretical return predicted by the CAPM.

Other methods evolved for measuring expected return as well. Ross (1976) creates the arbitrage pricing theory (APT), which calculates an asset's expected return based on its linear relationship with multiple macroeconomic factors that capture systemic risk. Although both CAPM and APT look primarily at market risk, the latter employs more factors. Fama and French (1993) expand the CAPM equation by adding a second size risk factor based on market capitalization and a third value risk factor based on the book-to-market ratio. Carhart (1997) adds momentum as a fourth contributing factor to explain the risk-return framework.

These concepts and many more are discussed further in this book. Readers of *Understanding Investment Risk and Return* will have the opportunity to gain an in-depth understanding of how investment risk can be defined, measured, and assessed. Likewise, readers will better understand the techniques available to measure returns, particularly over time. Combining risk and return allows for effective ways for investors to assess risk-adjusted performance. Finally, the overall conclusions of research studies enhance the reader's understanding as it is the main conceptual framework investors need to understand to apply such principles. Contributing authors are a mix of academics and practitioners.

ABOUT THIS BOOK

This section discusses the book's purpose, distinguishing features, intended audience, and structure.

Purpose

Understanding Investment Risk and Return surveys the tradeoff between risk and return and the value added that comes from a careful analysis of the interaction of the two. It highlights appropriate frameworks for measuring risk and returns while recognizing the impact of such tradeoffs over longer periods. Exploring the nature of differences in risk and return across asset classes and how to make modifications through derivative securities helps the reader to understand the integrated nature of risk and return more holistically. Not all investors view risk rationally, and such viewpoints impact micro and macro levels. *Understanding Investment Risk and Return* explores the consequences of past miscalculations of risk and returns while setting the stage for understanding how advanced techniques and emerging developments will help improve decision-making in the future. Rather than focusing on tedious statistical frameworks, this book presents the topics with a conceptual understanding of the findings of past work, bringing in mathematical concepts where necessary to highlight a broader understanding of topics related to risk and return.

Many books provide a theoretical, often technical, explanation of asset pricing models, including mathematical derivations, which are well suited to an academic or graduate-level content presentation. Likewise, many titles attempt a simplistic guide for beginner investors interested in understanding how risk and return intersect. This book seeks to bridge the gap between the two, bringing in research findings but doing so in a way that translates well for practitioners and individual investors looking for a deeper understanding of the topic without digging through unnecessary complex theoretical frameworks.

The scope of the coverage is broad and attempts to cover the most important aspects of risk and return management. The book consists of the following four sections.

- *Foundations of Risk and Return.* The first section provides an overview of how managing risk and return adds value, how to measure and manage risk, and how to measure and manage return.
- *Risk-Return Management and Analysis.* The second section focuses on managing risk and returns over time, risk-adjusted performance measurement, risk prioritization and understanding, applying, and critiquing different factor models.
- *Risk and Asset Classes.* The third section explains the risk-return differences across asset classes, how derivatives can be used to speculate on and hedge risk, and the role irrational investors play in risk-return tradeoffs.
- *Going Forward.* The fourth section focuses on the future by first examining lessons learned from past market bubbles and bankruptcies and emerging tools and models, including AI and machine learning.

Distinguishing Features

Understanding Investment Risk and Return has several distinguishing features.

- The book offers a synthesis of research studies in an easy-to-follow presentation. It provides a detailed presentation of investment risk and return practices. It is written to combine the conceptual world of scholars with the applied view of practitioners. The book contains contributions and perspectives from authors, including those from academia and practice.
- Results of empirical studies, when presented, link theory with practice in a concise manner that can be useful for sophisticated readers. Each chapter contains four to six discussion questions that reinforce key concepts discussed. Guideline answers are presented at the end of the book. This feature should be especially important to faculty and students using the book in the classroom.

Intended Audience

Our primary target audience for *Understanding Investment Risk and Return* comprises sophisticated practitioners, individual and institutional investors, academics, graduate finance students, and libraries. Experienced practitioners can use this book to navigate the key investment risk and return areas. Individual and institutional investors can expand their knowledge and apply the content to enhance portfolio management. Academics can use this book for a special topics course on risk or return or as a supplemental text for investment management courses. Graduate students can benefit from this book as a reference source. Libraries may find this book a useful addition to their collection. The book covers a broad array of topics that are relevant to investments, portfolio management, and risk management programs.

Book Structure

Understanding Investment Risk and Return consists of 12 chapters divided into four sections. The highlights of each section and chapter follow.

Part I: Foundations of Risk and Return

This section contains two other chapters in addition to the current chapter. Chapter 2 provides specific guidance about measuring and managing risk, and Chapter 3 provides guidance on measuring and managing return. Taken together, the chapters define risk and return and discuss how each is assessed.

Chapter 2: Measuring and managing risk (Xudong Fu)

Navigating finance's intricate and ever-changing landscape requires a deep understanding of risk and its management. This chapter thoroughly examines various types of risk, such as liquidity risk, regulatory risk, interest rate risk, reinvestment rate risk, price risk, volatility risk, correlation risk, credit (default) risk, counterparty risk, inflation risk, country/political/foreign exchange risk, and risk aggregation. Mastering risk measurement and management is essential for anyone engaged in finance and investing. Adopting a comprehensive and cohesive approach with both quantitative and qualitative methods to risk management allows investors to construct resilient and diversified portfolios capable of enduring the uncertainties and challenges of the financial industry.

Chapter 3: Measuring and managing returns (Timothy A. Krause, Eric J. Robbins, Santina A. Haddad)

This chapter explores the many ways to measure and manage returns. Individual investors can measure returns at a point in time or over multiple holding

periods. It is a good practice to consider returns after adjusting for inflation and the costs associated with the strategy. It is also important to include all sources of return, including capital appreciation and income (e.g., dividends or interest) received. Bond investors are concerned with income and capital appreciation (or lack of depreciation), but there are other nuances. These return factors include the coupon rate and relevant yield metrics like current yield, yield-to-maturity (YTM), yield-to-call (YTC), and yield-to-worst (YTW). Investors can consider earnings-based return measurements that can drive returns in financial markets. These include measurements like return on assets (ROA), return on equity (ROE), and return on invested capital (ROIC).

Part II: Risk-Return Management and Analysis

This part contains four chapters. Chapter 4 explores the impact that inflation has on risk and return. Chapter 5 introduces how to assess portfolio performance through risk-adjusted performance measures. Chapter 6 discusses how investors should prioritize the risks they take in their portfolios. Chapter 7 focuses on different factor models used for pricing assets and their role in the security and portfolio selection process.

Chapter 4: Risk and return across inflationary regimes (Jarah D. Macfarlane)

This chapter examines the dynamic relationship between risk, return, and inflation across various economic environments. It explores how correlation structures among assets change during deflationary, target inflation, and inflationary regimes, impacting diversification strategies and portfolio risk. The chapter discusses different types of inflation, their economic drivers, and policy implications. It analyzes products designed to reallocate risk automatically through asset allocation, focusing on their operation and suitability for different investors. It also reviews inflation-responsive products and how they aim to protect investors. Case studies from the past century, including the Great Depression, Japan's Lost Decade, 1970s stagflation, and the 2022 reopening shock, provide historical context and insights. The chapter also examines key inflation metrics and their incorporation into investment strategies. By understanding these relationships and employing appropriate investment approaches, investors can better manage their portfolios, achieve long-term financial goals, and potentially increase risk-adjusted returns in varying inflationary environments.

Chapter 5: Risk-adjusted performance measurement (Pilar Grau-Carles)

Investment performance measures play a critical role in evaluating the effectiveness of portfolios and asset managers. Key metrics, such as the Sharpe, Sortino, Treynor, and information ratios, assess risk-adjusted returns, focusing on performance relative to benchmarks or targets. The Sharpe ratio evaluates the excess return of an investment per unit of total risk, considering both upside and downside volatility. In contrast, the Sortino ratio emphasizes downside risk by using only the volatility of negative returns. The Treynor ratio measures returns earned more than what could have been earned on a risk-free investment per unit of market risk, providing insight into a portfolio's performance relative to systematic risk. The information ratio measures a manager's ability to generate excess returns adjusted for tracking error, reflecting active management performance. Understanding these metrics aids investors in selecting investments aligned with their risk tolerance and objectives.

Chapter 6: Prioritizing risks (Auke Plantinga)

This chapter discusses how to broaden risk management by including alternative approaches in addition to the investor's current investment management approach. Risk management is often tailored to the investor's strategies and policies. The proposed approach invites investors to use risk attribution as inspiration for improving the investment process with new strategies or research ideas. By including strategies outside the one being executed by the investor, it is possible to improve risk attribution and learn about hidden risks. The chapter illustrates the approach in the context of dedicated investment strategies with a hierarchical top-down investment process that focuses on policy decisions, market-timing decisions, and security-selection decisions. The analysis is based on a Monte Carlo simulation study.

Chapter 7: Understanding, applying, and critiquing different factor models (Rustin T. Yerkes)

Single/multi-factor asset pricing models are tools investors can use to select securities and build portfolios. These models fall at the intersection of the efficient frontier and investor objectives since asset prices ultimately reflect asset-specific features (timing, amount, and variability of future cash flows) and investor preferences. In another way, investors have risk preferences and return objectives and are faced with making trade-offs between these, subject to the available risk-return features offered by the universe of assets. Ultimately, an investor's portfolio choice can be modeled according to MPT as a mean-variance optimization problem where an investor seeks to maximize expected utility subject to an individual level of risk aversion. Expected utility is frequently modeled as a function of the expected value of an asset's excess return (the

return over and above the risk-free rate or risk premium) offset by a penalty for an investor's risk aversion.

Part III: Risk and Asset Classes

Section 3 contains three chapters. Chapter 8 explains the diversification benefits and higher returns associated with integrating digital assets, such as cryptocurrencies, into traditional multi-asset class portfolios. Chapter 9 discusses the roles that derivatives like forwards, futures, options, and swaps play in hedging, speculation, and portfolio management. Chapter 10 examines irrational investor behavior by exploring various behavioral tendencies, biases, and anomalies that shape investment decisions and market dynamics.

Chapter 8: Digital assets (cryptocurrencies): part of a multi-asset portfolio (Pankaj Agrawal and Doug Waggle)

This chapter explores the integration of digital assets (DAs), specifically cryptocurrencies, into traditional multi-asset class (MAC) portfolios, creating a DA-MAC framework. Well-constructed DA-MAC portfolios can offer significant diversification benefits and improved total returns compared to conventional MAC portfolios with equities, bonds, commodities, and real estate. However, due to their higher volatility, DAs should be limited to smaller proportions for most investor portfolios. The chapter applies established portfolio construction methodologies to incorporate cryptocurrencies, addressing market efficiency and price discovery issues. Empirical data supports the diversification benefits of DA-MAC portfolios, showing reduced cross-asset correlations and an upward shift in the efficient frontier. While DA-MAC portfolios present interesting opportunities, the role of cryptocurrencies in investment strategies remains a topic of ongoing discussion due to their volatility and regulatory concerns. The US government's recent approval of DA exchange traded futures (ETFs) marks a significant milestone in their integration into mainstream finance and the asset allocation process.

Chapter 9: Speculating return and hedging risk with derivatives (Tian (Lori) Tang)

Investors and businesses often face the dual challenge of seeking return opportunities while managing risks effectively in financial markets. Speculating returns and hedging risk with derivatives are two fundamental strategies to address these dual objectives. This chapter delves into the intricate world of speculating returns and hedging risks with derivatives, offering a comprehensive exploration of various derivative instruments, including forward contracts, futures contracts, options contracts, and swap contracts. The chapter elucidates the strategic application of derivatives in speculative endeavors and risk

management practices through detailed numerical examples and insightful case studies. The chapter also addresses regulatory considerations, best practices, and challenges in derivative trading, offering a comprehensive framework for investors and businesses to navigate derivative markets effectively.

Chapter 10: Irrational investors (Xing Lu)

This chapter delves into key behavioral tendencies, such as risk aversion and loss aversion, which explain why investors sometimes prioritize psychological comfort over objective financial analysis. It also examines various behavioral biases like overconfidence, herding behavior, anchoring, and confirmation bias, all contributing to suboptimal investment decisions. Additionally, the chapter discusses market anomalies like the momentum effect and the low-volatility anomaly, challenging the notion of market efficiency. Human error, including operational risks and cognitive mistakes, further exacerbates irrational decision-making. By understanding these elements of behavioral finance, investors and financial professionals can better navigate market dynamics and develop strategies to improve decision-making and investment outcomes.

Part IV: Going Forward

Part four contains two chapters. Chapter 11 investigates extreme outcomes in stock market returns, such as significant bull markets, bear markets, bubbles, and bankruptcies. Chapter 12 discusses future trends, tools, and techniques for measuring risk and return.

Chapter 11: Bulls, bears, bubbles, and bankruptcy: what have we learned? (Melissa Woodley)

This chapter examines extreme outcomes in returns and their causes. Investors who are students of market history may notice that stock prices tend to move in tandem, with the general flow of market returns moving in the same direction for extended periods. Occasionally, stock prices increase dramatically like bubbles, only to fall or pop equally dramatically. At other times, investors who ride a dramatic increase in a stock's price are rewarded as the stock maintains the new, higher value. One of the worst outcomes for investors is that firms also tend to file for corporate bankruptcy in waves, and shareholders of those companies often see their investments evaporate. This chapter explores the underlying causes of these market regimes and the latest evidence on how investors may identify signs of abrupt changes to the market landscape.

Chapter 12: Future trends and models for managing risk and return (M. Tony Via)

This chapter discusses emerging trends and research methodologies related to risk and return. Improvements in sensor technology in recent years and the ability to store data in digital formats have created a vast proliferation of machine-readable data worldwide. Coupled with the rapid increase in computing power in recent decades, the abundance of data is leading to the dominance of AI and machine learning methods over traditional factor modeling in financial research. These nascent developments have introduced a greater potential for unforeseen risks, especially as humans become more connected and behavioral science trends make more of an impact. New areas, such as “explainable AI,” and prompt engineering have risen to improve understanding of these systems. These innovations have expanded the need for certified experts in emerging subfields of financial risk, and the chapter closes with a discussion of risk-based financial certification agencies and programs.

SUMMARY AND CONCLUSIONS

This book is a comprehensive guide that explores the fundamental relationship between risk and return in modern investing. The book delves into sophisticated methods for measuring and managing risk across diverse asset classes, introducing readers to risk-adjusted performance metrics and advanced factor models. The chapters combine traditional financial theories with cutting-edge AI and machine-learning applications in investment analysis. The book examines various asset classes, including less traditional investments like digital assets, such as cryptocurrency and derivative-like options. Discussing these assets covers a spectrum of strategies, from conservative approaches like diversification and hedging to more speculative tactics. The book also explores why investors make irrational financial decisions by exploring behavioral biases such as risk tolerance and overconfidence. Finally, the book chronologically bridges historical examples of risk versus return to future considerations, including trends and expectations. Suitable for individual and institutional investors, the book provides an in-depth view of risk-return trade-offs, equipping readers with the tools to make informed investment decisions in an increasingly complex financial landscape.

REFERENCES

- Carhart, Mark M. 1997. “On Persistence in Mutual Fund Performance.” *Journal of Finance* 52(1), 57–82.
- Fama, Eugene F., and Kenneth R. French. 1993. “Common Risk Factors in the Returns on Stocks and Bonds.” *Journal of Financial Economics* 33(1), 3–56.

- Jensen, Michael C. 1967. "The Performance of Mutual Funds in the Period 1945–1964." *Journal of Finance* 23(2), 389–416.
- Jorion, Philippe. 2007. *Value-at-Risk: The New Benchmark for Managing Financial Risk*, 3rd Edition. New York: McGraw-Hill.
- Lintner, John. 1965. "The Valuation of Risky Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets." *Review of Economics and Statistics* 47(1), 13–37.
- Markowitz, Harry. 1952. "Portfolio Selection." *Journal of Finance* 7(1), 77–91.
- Ross, Stephen. 1976. "The Arbitrage Theory of Capital Asset Pricing." *Journal of Economic Theory* 13(3), 341–360.
- Sharpe, William. 1964. "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk." *Journal of Finance* 19(3), 425–442.
- Sharpe, William. 1966. "Mutual Fund Performance." *Journal of Business* 39(1), 119–138.
- Sharpe, William. 1994. "The Sharpe Ratio." *Journal of Portfolio Management* 21(1), 49–58.
- Sortino, Frank A., and Lee N. Price. 1994. "Performance Measurement in a Downside Risk Framework." *Journal of Investing* 3(3), 54–69.
- Treynor, Jack L. 1966. "How to Rate Management Investment Funds." *Harvard Business Review* 43(1), 63–75.